

Q- Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.

To configure a Site-to-Site VPN in Cisco Packet Tracer so that two different branch networks can communicate securely through a VPN tunnel using GRE, IPsec, and routing.

2. Network Topology

PC1 — Switch1 — Router0 **=====** Router1 — Switch2 — PC2
VPN Tunnel

3. IP Addressing

Device	Interface	IP Address	Subnet Mask
Router0	G0/0	192.168.1.1	255.255.255.0
Router0	G0/1	10.0.0.1	255.255.255.252
Router1	G0/1	10.0.0.2	255.255.255.252
Router1	G0/0	192.168.2.1	255.255.255.0
Router0	Tunnel0	172.16.0.1	255.255.255.252
Router1	Tunnel0	172.16.0.2	255.255.255.252
PC1	Fa0	192.168.1.10	255.255.255.0
PC2	Fa0	192.168.2.10	255.255.255.0

4. Router0 – Interface Configuration

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
```

5. Router1 – Interface Configuration

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.0.2 255.255.255.252
```

```
no shutdown
exit
```

6. GRE Tunnel – Router0

```
interface tunnel 0
ip address 172.16.0.1 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 10.0.0.2
no shutdown
exit
```

7. GRE Tunnel – Router1

```
interface tunnel 0
ip address 172.16.0.2 255.255.255.252
tunnel source gigabitEthernet 0/1
tunnel destination 10.0.0.1
no shutdown
exit
```

8. Static Routing – Router0

```
ip route 192.168.2.0 255.255.255.0 172.16.0.2
```

9. Static Routing – Router1

```
ip route 192.168.1.0 255.255.255.0 172.16.0.1
```

10. IPsec Configuration – Router0

```
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
exit
```

```
crypto isakmp key VPNKEY address 10.0.0.2
```

```
crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac
```

```
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
```

```
crypto map VPN_MAP 10 ipsec-isakmp
set peer 10.0.0.2
set transform-set VPN_SET
match address 100
exit
```

```
interface gigabitEthernet 0/1
crypto map VPN_MAP
exit
```

11. IPsec Configuration – Router1

```
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
```

```

group 2
exit

crypto isakmp key VPNKEY address 10.0.0.1

crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac

access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255

crypto map VPN_MAP 10 ipsec-isakmp
set peer 10.0.0.1
set transform-set VPN_SET
match address 100
exit

interface gigabitEthernet 0/1
crypto map VPN_MAP
exit

```

12. PC1 Configuration

IP Address: 192.168.1.10
 Subnet Mask: 255.255.255.0
 Default Gateway: 192.168.1.1

13. PC2 Configuration

IP Address: 192.168.2.10
 Subnet Mask: 255.255.255.0
 Default Gateway: 192.168.2.1

14. Verification Commands

```

show interfaces tunnel 0
show ip route
show crypto isakmp sa
show crypto ipsec sa
show crypto map

```

15. Connectivity Test

From PC1:
 ping 192.168.2.10

From PC2:
 ping 192.168.1.10

A Site-to-Site VPN was configured between Router0 and Router1 using a GRE tunnel protected by IPsec. Static routes were added so that the two branch LANs could communicate through the VPN. The configuration can be verified using tunnel, routing, ISAKMP, IPsec, and end-to-end ping commands.